Department of Information Technology

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch : V Semester-B. Tech Information Technology
Course Code & Title : CS3591 Computer Networks
Name of the Faculty member (s) : Mrs.M.Thulasi devi, AP/IT

Innovative Practice Description

- **Unit / Topic : Unit 4 / Routing Protocols, Unicast Protocol**
- **Course Outcome :** CO4
- **Topic Learning Outcome:** TLO14
- **Activity Chosen : Think Pair Share**
- **Justification:**
  - The "Think-Pair-Share" (TPS) instructional approach proves exceptionally suitable for intricate subjects such as routing protocols and unicast protocols in computer networking, owing to multiple critical elements that correspond with the characteristics of these topics and learner requirements.
  - This methodology is optimal for instructing Routing Protocols and Unicast Protocols as it dynamically involves students in analytical reasoning, teamwork, and solution development, which are fundamental for comprehending these sophisticated concepts.
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
  - Instructor explained introducing a specific scenario or problem related to routing, such as determining the best path for a unicast packet in a network using a particular protocol (e.g., OSPF or RIP).
  - The instructor presented a particular case study or challenge concerning routing, such as identifying the optimal pathway for a unicast packet within a network utilizing a specific protocol (e.g., OSPF or RIP).
  - Initially, students were provided 3-5 minutes to **contemplate** independently regarding the elements influencing routing choices, including link expense, hop distance, or network structure as demonstrated in Figure 1.
  - Subsequently, students were instructed to **collaborate** in pairs and exchange their perspectives for 5-7 minutes, motivating them to contrast their logic, resolve uncertainties, and enhance their comprehension through collaborative learning as illustrated in Figure 2.
  - Ultimately, selected students **presented** their conclusions, with each partnership offering

perspectives on routing decision-making processes, the function of unicast addresses, or the advantages and constraints of the selected protocol as depicted in Figure 3.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	3	3	2	2	3	1	2	2	1	2	1	1	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
	3	3	2	2	3	1	2	2	1	2	1	1	3
Justification for correlation	The students could be able to use and apply basic concept of transport layer protocol	The students could be able to review the real time problem with the transport layer protocol	The students will be able to address various problems related to transport layer	The students will be capable of understanding the various uncertainties in transport layer	The students will be capable of developing or applying various frameworks for managing transport layer	In practice, the students will adhere to ethics in various protocols in transport layer	The students will work as part of a team and individually to illustrate their competencies in the transport layer	The students will illustrate their abilities in communication and problem-solving as they engage with the transport layer	The students will be able to identify the task required to build TCP and UDP	The student will identify the technological advances made in the past and update the concepts in transport layer	The students could be able to apply computing skills to the transport layer protocols.	The students will have the ability to provide IT solutions concerning the transport layer.	The students could be able to develop software application for transport layer protocols

- **Images / Screenshot of the practice**

Figure 1. Pair with neighbours and discuss answer

Figure 2. S Poongothai share the answer to entire class

- **Reflective Critique:**

  - ***Feedback of practice from students and other stakeholders:***
    - Participants indicated that the exercise proved highly beneficial for implementing the principles they acquired to address the presented challenge effectively.
    - Participants mentioned that it will provide significant assistance during examination preparation.
    - They experienced ease in conveying their comprehension of the concept with their classmates.

  - ***Benefit of the practice:***
    - The Think-Pair-Share exercise provides learners with occasions to exchange their conceptual understanding with colleagues and simultaneously enhances their verbal expression and comprehension abilities.
    - This exercise demonstrated strong interactivity and enabled struggling learners to address their questions with their classmates.
    - It contributes to advancing students' verbal communication proficiency by articulating the principles they absorbed during instruction.

  - ***Challenges faced in implementation:***
    - Limited student partnerships were arbitrarily chosen to deliver their presentations before the class owing to temporal limitations.
    - Ensuring students recognize the value and importance of articulating their conceptual understanding.
    - The exercise requires extended duration for participants to express their perspectives.

References:

- https://www.ritrjpm.ac.in/images/computer-science/2021-2022/3.VA_OCS752-ThinkPairShare.pdf
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/TPS-RV-I-IT-PSPP.pdf
- https://www.readingrockets.org/strategies/think-pair-share

Signature of Faculty Member **HOD**